

Blockchain: Helping Industries Grow

Blockchain technology, once known primarily as the backbone of cryptocurrencies, has evolved into a transformative force across various industries. This comprehensive guide explores the diverse ways blockchain is revolutionising industries worldwide.



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Understanding the blockchain's key technology is crucial as it forms the foundation of ensuring data security and allowing for decentralised consensus. Table 1 explains the key terminologies of the blockchain that need to be understood by the user.

How does blockchain technology work?

Blockchain technology organises data into blocks interconnected in chains, each containing a hash

from the preceding one, a time stamp, and a digital signature of the transaction data. Before a transaction can be incorporated into the current blockchain, it undergoes several crucial steps, as illustrated in Figure 1.

Step 1: User 1 initiates the transaction to send X amount of digital cryptocurrency to User 2.

Step 2: User 1's node begins the transaction by creating it and then digitally signing it with its private key. Transactions within a blockchain can represent various actions.

Step 3: The requested transaction is disseminated across a peer-to-peer (P2P) network and broadcast to every computer or node.

Step 4: Each node receives the request and endeavours to validate the transaction using cryptographic algorithms. The node that completes the validation process is known as the miner node, which is rewarded with cryptocurrency.

Step 5: The validated blocks are then appended to the existing blockchain network. These nodes